

Pythagoras' Theorem & Trigonometry

Extra practice test 4 — answer sheet · 27 marks total

Section A — Pythagoras' theorem

A1. $6^2 + 6^2 = 36 + 36 = 72$ (1) hypotenuse = $\sqrt{72} = 6\sqrt{2} \approx 8.5$ cm (1)

A2. $a^2 = 10^2 - 7^2$ (1) $= 100 - 49 = 51$ (1) $a = \sqrt{51} = 7.1$ cm (1)

A3. $\text{length}^2 = 4.8^2 + 2.2^2 = 23.04 + 4.84 = 27.88$ (1) $\text{length} = \sqrt{27.88} = 5.3$ m (1)

Section B — SOH CAH TOA basics

B1. SOH CAH TOA — specifically **CAH** (adjacent = hypotenuse $\times \cos \theta$) (1)

B2. **Pythagoras' theorem** — no angle is known or needed (1)

B3. **False** — Pythagoras' theorem only applies to right-angled triangles (1)

Section C — Finding missing sides

C1. $\sin 22^\circ = 6 \div h$ (1) $h = 6 \div \sin 22^\circ$ (1) $= 16.0$ cm (1)

C2. $\cos 66^\circ = x \div 9 \rightarrow x = 9 \times \cos 66^\circ$ (1) $= 3.7$ cm (1)

C3. $\tan 39^\circ = 7 \div x \rightarrow x = 7 \div \tan 39^\circ$ (1) $= 8.6$ cm (1)

Section D — Finding missing angles

D1. $\cos \theta = 7 \div 25 = 0.28$ (1) $\theta = \cos^{-1}(0.28) = 73.7^\circ$ (1)

D2. $\tan \theta = 14 \div 40 = 0.35$ (1) $\theta = \tan^{-1}(0.35)$ (1) $= 19.3^\circ$ (1)

Section E — Applying Pythagoras & trigonometry

E1. (a) $\text{hyp}^2 = 9^2 + 12^2 = 81 + 144 = 225 \rightarrow \text{hyp} = 15$ cm (1) (b) $\sin \theta = 9 \div 15 = 0.6 \rightarrow \theta = \sin^{-1}(0.6) = 36.9^\circ$ (1)

E2. (a) $\text{strut}^2 = 1.8^2 + 3^2 = 3.24 + 9 = 12.24 \rightarrow \text{strut} = \sqrt{12.24} = 3.5$ m (1) (b) $\tan \theta = 1.8 \div 3 = 0.6$ (1) $\theta = \tan^{-1}(0.6) = 31.0^\circ$ (1)

Marking note: award method marks for a correctly set-up equation or calculation even if the final answer contains a rounding or arithmetic slip. Accept any correct rounding to 1 decimal place unless a different level of accuracy is specified. Missing units on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.